

Manav Patel

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PROFESSIONAL SUMMARY

Data engineer passionate about building scalable intelligent systems and data-driven solutions. Skilled in Python, SQL, Apache Airflow, and cloud platforms. Experienced in designing end-to-end pipelines from data ingestion to predictive analytics and visualization.

EDUCATION

San Jose State University	Aug 2025 – May 2027
Master's in Applied Data Intelligence (GPA: 3.8/4.0)	
Navrachna University	Sept 2022 – May 2025
B.Tech in Computer Science Engineering (GPA: 3.63/4.0)	

EXPERIENCE

TactTree | Data Analytics & Business Intelligence Intern Jan 2025 – June 2025

- Built automated **SQL** and **Python** ETL pipelines for multi-source data cleaning and validation, establishing reliable data workflows across operational systems.
- Engineered executive-level **Tableau** and **Power BI** dashboards integrating advanced statistical forecasting and predictive modeling, enabling leadership to make data-informed strategic decisions.
- Conducted longitudinal statistical analysis on large-scale datasets using **Pandas** and **Scikit-learn**, identifying emerging market trends and providing actionable insights.
- Developed custom **Python** scripts and automated reporting workflows, streamlining complex analytical processes and reducing manual operational overhead.

Rishabh Software | AI/ML Engineer Intern Jun 2024 – Dec 2024

- Architected an enterprise **RAG system** using **LangChain** and **Pinecone** to automate semantic search and information discovery across document repositories, enabling efficient knowledge retrieval at scale.
- Fine-tuned **Large Language Models (LLMs)** via RLHF and instruction-tuning to enhance intent classification and contextual accuracy for sentiment analysis; validated improvements through rigorous testing.
- Engineered a production-grade predictive maintenance engine using **LSTM** and **XGBoost** architectures for real-time anomaly detection on industrial telemetry datasets, improving fault detection reliability.
- Operationalized ML workflows by implementing **MLOps** pipelines with **Docker** and **CI/CD**, establishing scalable deployment processes and ensuring reproducibility from research to production environments.

TECHNICAL SKILLS

Technical Skills	Data Engineering, Data Analysis, Machine Learning, Feature Engineering, Generative AI, Cloud Infrastructure
AI/ML Algorithms	Random Forest, XGBoost, LangGraph, LSTM, CNN, Transfer Learning, RLHF, RAG Systems, LLM Fine-tuning, Computer Vision, NLP
Programming Languages	Python, SQL, Java, C++, Dart, JavaScript, HTML/CSS, Matlab
Python Libraries	Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost, YOLO, OpenCV, Hugging Face, LangChain, Pinecone
Data Infrastructure	Apache Airflow, Snowflake, dbt, Docker, AWS (S3, EC2, SageMaker, ECS), PostgreSQL, BigQuery, CI/CD Pipelines
Generative AI	LLM APIs (OpenAI, Azure), Prompt Engineering, LangChain Integration, LLM Evaluation, AI Orchestration with LangGraph, Ollama
Visualization Tools	Tableau, Power BI, Streamlit, Matplotlib, Seaborn, Plotly
Frameworks & Platforms	FastAPI, React, Supabase, Firebase, MLflow, Kubernetes
Development Tools	VS Code, Jupyter Notebook, Google Colab, Postman, Git, GitHub, GPU Acceleration, Docker

PROJECTS

Agentic AI Systems & Distributed Workflows | LangGraph, FastAPI, Cloud Infrastructure 2026

- Engineered multi-step autonomous AI workflows using **LangGraph's** Planner-Reviewer patterns with stateful **AgentState** management, enabling self-correcting reasoning loops and conditional task routing.
- Built deterministic state management with **JSON schema validation** and integrated **LLM APIs** (OpenAI, **Ollama**) with modular prompt pipelines supporting seamless model switching.
- Implemented structured error-handling loops with intelligent feedback mechanisms enabling autonomous self-correction and resilience across iterative reasoning workflows.
- Deployed full-stack applications using **Docker** and **AWS ECS** across multi-component systems (**React**, **Python LangChain**, **FastAPI**); benchmarked RAG systems using **LlamaIndex** semantic chunking.

- Engineered end-to-end data pipeline infrastructure using **Apache Airflow**, orchestrating 4 DAGs (BLS, Adzuna, College Scorecard, WARN) that extract employment, job market, and education data into **Snowflake** warehouse.
- Developed **dbt** transformation models (staging + mart layer) that standardize data from multiple sources into a unified analytical schema, enabling cross-functional analysis of graduate employment outcomes and degree ROI.
- Designed scalable **Snowflake** architecture with RAW/ANALYTICS/SNAPSHOTS schemas, implementing MERGE-based upserts, batch processing (1K rows/batch), and historical snapshots for institution outcomes tracking.
- Built AI-driven career intelligence system using **OpenAI API**, creating an intelligent chatbot for personalized career path recommendations, job-degree relevance scoring, and 10-year employment forecasting.

Global Weather & Air Quality Predictive Analytics System | *Python, Scikit-learn, XGBoost, Tableau*

2025

- Engineered end-to-end machine learning system processing 109,718 weather records with production-grade data pipelines: implemented country-stratified imputation, statistical outlier management via winsorization, and ordinal encoding schemes for cyclical variables.
- Designed feature transformation architecture extracting temporal decomposition (hour, minute), timestamp conversions, and cyclical feature representations for wind direction and lunar phases to capture non-linear environmental dependencies.
- Trained and evaluated three regression models (**Random Forest**, **XGBoost**, **K-NN**) using **ColumnTransformer** preprocessing; Random Forest achieved exceptional generalization across temperature, PM2.5, and AQI predictions on unseen test data.
- Deployed interactive **Tableau** dashboard synthesizing historical and predictive analytics with geospatial risk mapping, country-level scenario forecasting, and temporal heatmaps supporting public health policy and environmental governance decisions.

TapEats - Food Ordering System | *Flutter, Supabase, Dart*

2025

- Engineered cross-platform mobile application with real-time ordering, QR menu integration, and push notifications; implemented complete customer workflow from menu browsing to order confirmation.
- Architected **Supabase** real-time database backend with PostgreSQL, enabling synchronous data updates across multiple user sessions and instant order notifications.
- Built admin dashboard for restaurant operations including sales tracking, inventory management, and order status updates; demonstrated full-stack understanding from user interface to database architecture.

KHOJ: Flood Recovery Financial Intelligence | *Python, Tableau, Power BI, Statistics*

2024

- Spearheaded a data initiative to quantify financial burdens of flood recovery by processing raw field data from 370+ households.
- Architected high-fidelity **Tableau** and **Power BI** dashboards to translate fragmented datasets into actionable intelligence.
- Performed empirical evaluation and error analysis to improve robustness of data findings across different demographics.

CERTIFICATIONS

- AWS Academy Graduate – Machine Learning, NLP, Cloud Foundations & Security (2024)
- Google Data Analytics Professional Certificate (2024)
- Python for Data Science and AI – IBM (2024)